

NAYAN DHURVE

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Summary

Machine Learning Engineer building AI-powered decision-support systems that combine machine learning, explainable AI, and data engineering to solve real business problems. Experienced in delivering end-to-end pipelines on HR and banking datasets from ETL and feature engineering to model development, evaluation, and business intelligence integration. Translating data into actionable decisions through rigorous statistical analysis, model interpretability, and cross-functional communication.

Technical Skills

Languages: Python · SQL · C

ML & DL: Scikit-learn · TensorFlow · Keras · XGBoost · SHAP · Prophet · SMOTE

Data Engineering: Pandas · NumPy · ETL · Feature Engineering · Statistical Imputation

Statistics & BI: Kaplan-Meier · Cox PH · ANOVA · Chi-square · Power BI · Plotly · Matplotlib

Generative AI: LLMs · Prompt Engineering · RAG · Google AI Studio

Engineering: Git · GitHub · REST APIs · pytest · CI/CD · Render · Vercel

Experience

Data Analytics Intern

Jan 2025 – Jul 2025

ICEICO Technologies Pvt. Ltd.

Nagpur, India

- Engineered an end-to-end HR analytics platform transforming 3,000+ employee records into a predictive decision-support system spanning ETL, feature engineering, statistical modeling, and BI delivery.
- Built a 7-stage ETL pipeline with statistical imputation and engineered 53 analytical features, transforming raw HR data into structured datasets for attrition prediction.
- Developed explainable attrition models using XGBoost (86.7% accuracy) and Logistic Regression (0.851 ROC-AUC, 96.9% recall), selecting Logistic Regression over XGBoost for HR compliance due to complete SHAP-based interpretability.
- Conducted survival analysis (Kaplan-Meier, Cox PH), ANOVA, and Chi-square tests to identify statistically significant attrition factors, uncovering a high-risk segment with 36.8% termination rate, 3× above baseline, and 49 high-potential employees for strategic retention.
- Developed executive Power BI dashboards and automated HTML reporting with 112 test cases, enabling actionable workforce KPIs for data-driven HR decision-making.

Projects

FinSight | AI-Powered Banking Intelligence Platform

[GitHub](#)

Tech: Python · TensorFlow · XGBoost · Prophet · Flask · Plotly · SQLite

Problem: Traditional banking systems lack predictive analytics and intelligent decision support for financial operations.

- Engineered an AI-powered banking intelligence platform using real RBI data across 65 banks, integrating forecasting, fraud detection, customer analytics, and ATM operations.
- Developed production-oriented ML pipelines for cash demand forecasting (Prophet), hybrid fraud detection (rule-based + Isolation Forest), churn prediction, and credit scoring through a unified model interface with consistent evaluation.
- Designed a versioned feature store enabling reproducible training and automated retraining across three specialized databases, ensuring consistent feature reuse across forecasting, fraud detection, and recommendation models.
- Established ML lifecycle practices including model monitoring, scheduled retraining, and 60+ automated tests for maintainable, reproducible analytics.

CosmoGuide | Multi-Provider LLM Platform

[GitHub](#) · [Live Demo](#)

Tech: Express.js · React · Gemini · Claude · Groq · NASA APIs

Problem: General AI assistants lack real-time space data and flexible model selection for astronomy and space exploration.

- Architected a multi-provider LLM gateway integrating 11 AI providers and 100+ models with unified routing, key management, failover, and configurable prompt strategies.
 - Integrated scientific visualization, live telemetry, and generative AI through five NASA APIs with resilient fallback mechanisms.
 - Designed a stateless backend with browser-based BYOK authentication and secure API proxying, supporting flexible provider selection without server-side storage.
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Education

B.Tech in Information Technology

2021 – 2025

Yeshwantrao Chavan College of Engineering, Nagpur

Certifications

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|---|----------|
| ▪ IBM (Coursera) — Python for Data Science, AI & Development | Jun 2023 |
| ▪ freeCodeCamp — Machine Learning with Python | Jun 2024 |
| ▪ Tata (Forage) — GenAI Powered Data Analytics Job Simulation | Dec 2025 |